

COMP 5350/6350 Hands-On #4 Report

Student Name: Eristen Forsythe

1. Executive Summary

Investigators analyzed six image files and two PDF documents to find hidden data, metadata issues, and possible embedded content. From the images, they recovered four map fragments of Auburn and four passphrases using steganography tools. In the documents, they discovered an encrypted ZIP file hidden inside SANS_Digital_Forensic.pdf, which was manually extracted using the password auburn2026.

The extracted files included a decoded message, a hidden spreadsheet-style file, an image of a getaway vehicle, and images of valuable targets. Altogether, the evidence points to a planned theft in Auburn, Alabama, with a meetup at Aurelia Bank at 12 noon, a green Ford Mustang (EP68 BLZ) as the getaway car, and a team with defined roles and profit shares.

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Questions	Answers
Which files contained which map fragment?	Part A = beach.jpg (ToomersCorner); Part B = nyc.jpg (TigerWalk7); Part C = lena.bmp (WarEagle2026); Part D = mountain.jpg (Shelby7).
What are the decoded messages contained within the images/documents?	"Lets gather up at Aurelia bank, Auburn at 12 noon. Lets go team!"
Which city is the target of the operation?	Auburn, Alabama.
What is the specific location and time of the planned event?	Aurelia Bank, Auburn at 12 noon.
What is the getaway vehicle model, color, and license plate?	Green Ford Mustang, license plate EP68 BLZ.
Who are the individuals involved, and what are their roles or profit shares?	Cipher - Leader - 40%; Phantom - Safe Cracker - 25%; Byte - Getaway Driver - 15%; Sentinel - Lookout - 20%.
What is the target item of the heist?	The archive contained two high-value item references. The Hope Diamond appears to be the most likely primary target, with a Rolex Daytona 116506 Platinum "Platona" Baguette present as an additional target reference.
What password(s) were used and what files were they used for?	ToomersCorner = beach.jpg; WarEagle2026 = lena.bmp; Shelby7 = mountain.jpg; TigerWalk7 = nyc.jpg; auburn2026 = - Daytona_116506_Platinum_Platona_Baguette.jpg, - file.pdf, - Hope_Diamond.jpg, - EP68BLZ.jpg, - flag.txt.

2. Investigation Methodology (General Overview)

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The investigation used a repeatable workflow using evidence inventory, file identification, metadata review, file structure inspection, steganographic extraction, manual archive carving, and post-extraction interpretation. Work was documented through terminal captures, recovered artifact images, and annotated figures embedded throughout this report and the appendices

Methods	How
Evidence intake and inventory	Recovered image files, recovered PDF documents, and the provided passphrases.txt wordlist were staged in the analyst working directory and enumerated with ls, unzip, and find.
Integrity verification	SHA-256 hashes were generated for the six image files, the two PDF files, and the supporting passphrases.txt file to preserve a reference baseline and support reproducibility.
File-type and metadata review	file and exiftool were used to verify real file types, dimensions, and metadata fields. strings was used to surface embedded text or XMP metadata, especially in MAP.jpg.
Low-level structure inspection	xxd verified PNG structure for AU.png and inspected byte-level evidence for hidden or appended content. strings and xxd on SANS_Digital_Forensic.pdf exposed ZIP filenames and signatures near the end of the PDF.
Steganographic extraction	steghide info tested likely carrier files, and stegseek with passphrases.txt recovered four valid passphrases and extracted four hidden .out image fragments from beach.jpg, lena.bmp, mountain.jpg, and nyc.jpg.
Fragment interpretation	The recovered .out files were identified as JPEG image data and compared against the supplied 2×2 map grid to reconstruct the larger Auburn map.
Archive carving and extraction	binwalk identified an encrypted ZIP archive appended inside SANS_Digital_Forensic.pdf. Because automated extraction was incomplete, dd carved the archive at the reported offset. The archive was then listed and extracted manually.
Password usage and decoding	The carved hidden_archive.zip was unlocked with auburn2026. The recovered flag.txt content appeared as hexadecimal text and was decoded from hex to ASCII. The disguised file.pdf was identified as an OLE/Excel-style document and interpreted with file and strings.
Documentation approach	Key terminal outputs were preserved as screenshots, while recovered fragment images, vehicle imagery, and target-reference images were saved as evidentiary figures. The strongest screenshots were placed inline in the findings sections, with supplemental command captures in Appendix A.

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3. Evidence Overview

3.1 Evidence File Properties

File	Type	SHA-256
AU.png	PNG	6d9438a0c003115fc95eefede959e0b6f7b8905aaae7d18a38a7376775c83d04
beach.jpg	JPEG	b7d2bf1638d080fd4bf684864652b57cab96b02c542e068b28a5d8c8d331311d
lena.bmp	BMP	1c9b3006e5e6c4c2844e688a7aa5052420e8b0ee0aac11c7201297fb231f094b
MAP.jpg	JPEG	96bc51f94cf5414ca0d7537730d11b453fb412678753a40d69d61b0221e660a1
mountain.jpg	JPEG	a70de3b24740dc8dd40b6d1b99f4bfb292db747506088b766c2a927a06aaf01a
nyc.jpg	JPG	dd318f588c40ade466519307ea507ba3275f909b12a9c68fa6f3100d7d5232db
BAFLE-DCT.pdf	PDF	bdc8b6c1bba33b1077ce3da05b70e647077ef9ccacecfa341fd62ebf8ab08174
SANS_Digital_Forensic.pdf	PDF	80711312bc19c9ec0a3683dc99f66d1cee28038b7ae306ffe3782ff407079a3f
passphrases.txt	TXT	1e3f7c4b8eb11cc9fcae6412349cb41aef22a5268417833129798f0ebe42a949

3.2 Directory Overview

Recovered Images contained six files: AU.png, beach.jpg, lena.bmp, MAP.jpg, mountain.jpg, and nyc.jpg.

Their purposes were to be either a steganographic carrier file or visual context for map reconstruction. Recovered Documents contained two PDF files: BAFLE-DCT.pdf and SANS_Digital_Forensic.pdf.

The first appeared context-related, while the second concealed the archive that showed the operational details. Initial anomalies included image side passphrase recovery, extracted .out image fragments, suspicious XMP metadata in MAP.jpg, and ZIP indicators embedded in SANS_Digital_Forensic.pdf

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4. Evidence and Findings

4.1 Image Analysis and Hidden Data Extraction

This section follows the template field for field. Each recovered /supporting image is documented with file type, tools, commands, output, extracted content, passphrases, and the screenshots that support the finding

beach.jpg

File Name:	beach.jpg
File Type:	JPEG image file
Tool(s) Used:	file, steghide info, stegseek -wl passphrases.txt, file (for beach.jpg.out)
Command Used:	file /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/beach.jpg steghide info /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/beach.jpg stegseek -wl /home/EristenForsythe/Downloads/Recovered_items/passphrases.txt /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/beach.jpg file beach.jpg.out lena.bmp.out mountain.jpg.out nyc.jpg.out
Output:	stegseek recovered the passphrase “ToomersCorner” and extracted beach.jpg.out. beach.jpg.out was identified as JPEG image data (1228 × 696).
Extracted Content:	Northwest Auburn map fragment later assigned to Part A of the 2×2 reconstruction.
Password/Passphrase (if applicable):	ToomersCorner



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lena.bmp

File Name:	lena.bmp
File Type:	BMP image file
Tool(s) Used:	file, steghide info, stegseek -wl passphrases.txt, file (for lena.bmp.out)
Command Used:	file /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/lena.bmp steghide info /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/lena.bmp stegseek -wl /home/EristenForsythe/Downloads/Recovered_items/passphrases.txt /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/lena.bmp file beach.jpg.out lena.bmp.out mountain.jpg.out nyc.jpg.out
Output:	stegseek recovered the passphrase “WarEagle2026” and extracted lena.bmp.out. lena.bmp.out was identified as JPEG image data (1228 × 696).
Extracted Content:	Southwest Auburn map fragment later assigned to Part C of the 2×2 reconstruction.
Password/Passphrase (if applicable):	WarEagle2026



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mountain.jpg

File Name:	mountain.jpg
File Type:	JPEG image file
Tool(s) Used:	file, steghide info, stegseek -wl passphrases.txt, file (for mountain.jpg.out)
Command Used:	file /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/mountain.jpg steghide info /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/mountain.jpg stegseek -wl /home/EristenForsythe/Downloads/Recovered_items/passphrases.txt /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/mountain.jpg file beach.jpg.out lena.bmp.out mountain.jpg.out nyc.jpg.out
Output:	stegseek recovered the passphrase “Shelby7” and extracted mountain.jpg.out. mountain.jpg.out was identified as JPEG image data (1228 × 696).
Extracted Content:	Southeast Auburn map fragment later assigned to Part D of the 2×2 reconstruction.
Password/Passphrase (if applicable):	Shelby7



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nyc.jpg

File Name:	nyc.jpg
File Type:	JPEG image file
Tool(s) Used:	file, steghide info, stegseek -wl passphrases.txt, file (for nyc.jpg.out)
Command Used:	file /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/nyc.jpg steghide info /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/nyc.jpg stegseek -wl /home/EristenForsythe/Downloads/Recovered_items/passphrases.txt /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/nyc.jpg file beach.jpg.out lena.bmp.out mountain.jpg.out nyc.jpg.out
Output:	stegseek recovered the passphrase “TigerWalk7” and extracted nyc.jpg.out. nyc.jpg.out was identified as JPEG image data (1228 × 696).
Extracted Content:	Northeast Auburn map fragment later assigned to Part B of the 2×2 reconstruction.
Password/Passphrase (if applicable):	TigerWalk7

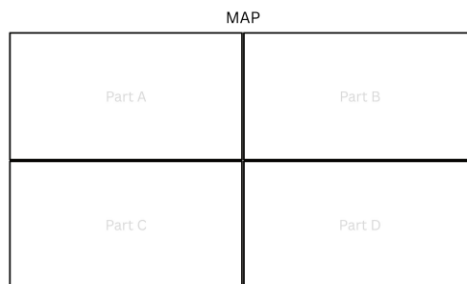


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MAP.jpg

File Name:	MAP.jpg
File Type:	JPEG image file
Tool(s) Used:	file, exiftool, strings, stegseek -wl passphrases.txt
Command Used:	<pre>file /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/MAP.jpg exiftool /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/MAP.jpg strings /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/MAP.jpg grep -E "xmpmeta Attrib Untitled" stegseek -wl /home/EristenForsythe/Downloads/Recovered_items/passphrases.txt /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/MAP.jpg</pre>
Output:	Metadata review exposed XMP and attribution fields, including “Untitled design - 1.” stegseek did not recover a valid passphrase from MAP.jpg.
Extracted Content:	No hidden fragment was recovered directly. The file served as contextual map-related evidence.
Password/Passphrase (if applicable):	No valid passphrase recovered



AU.png

File Name:	AU.png
File Type:	PNG image file
Tool(s) Used:	file, exiftool, xxd

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Command Used:	<pre>file /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/AU.png exiftool /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/AU.png xxd /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/AU.png head -20</pre>
Output:	AU.png was confirmed as a normal PNG with expected IHDR, PLTE, and IDAT chunks.
Extracted Content:	No hidden fragment or archive was recovered from AU.png. It remained supporting context only.
Password/Passphrase (if applicable):	Not applicable



Screenshots:

- shows file-type identification for the recovered evidence.
- shows MAP.jpg metadata grep output and AU.png byte-level structure.
- shows the recovered .out files identified as JPEG fragments and opened for visual review.

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Recorded terminal results for passphrase recovery are reproduced below :

StegSeek 0.6

[i] Found passphrase: "ToomersCorner" = beach.jpg.out

[i] Found passphrase: "WarEagle2026" = lena.bmp.out

[i] Found passphrase: "Shelby7" = mountain.jpg.out

[i] Found passphrase: "TigerWalk7" = nyc.jpg.out

[!] MAP.jpg: Could not find a valid passphrase.

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```
(EristenForsythe@digitalforensics) [~/Downloads/Recovered_items]
$ strings /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/MAP.jpg | grep -E 'xmpmeta|Attrib|Untitled'
<x:xmpmeta xmlns:x='adobe:meta:'>
  xmlns:Attrib='http://ns.attribution.com/ads/1.0/'
  <Attrib:Ads>
    <Attrib:Created>2025-11-12</Attrib:Created>
    <Attrib:ExtId>510c56c0-48b3-4603-8634-f91dab8d12ba</Attrib:ExtId>
    <Attrib:FbId>525265914179580</Attrib:FbId>
    <Attrib:TouchType>2</Attrib:TouchType>
  </Attrib:Ads>
  <rdf:li xml:lang='x-default'>Untitled design - 1</rdf:li>
</x:xmpmeta>

(EristenForsythe@digitalforensics) [~/Downloads/Recovered_items]
$ xxd /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/AU.png | head -20
00000000: 8950 4e47 0d0a 1a0a 0000 000d 4948 4452  PNG.....IHDR
00000010: 0000 00e1 0000 00e1 0803 0000 0009 6d22  .....m"
00000020: 4800 0001 5050 4c54 45ff ffff 0029 5b00  H...PPLTE....[.
00000030: 285b 0029 5c00 2757 0021 5501 2a5f 0027  ([.)\.'W.U.*_.'
00000040: 59c8 7747 242e 51ee 842d 0026 5cee 7f32  Y.wG$.Q..-.6\..2
00000050: e582 4053 4146 0024 5a48 3f46 f8c9 abef  ..@SAF.$ZH?F....
00000060: 7d2c f78b 3ae0 8244 4640 5145 414c 0023  },...DF@QEAL.#
00000070: 5c01 2956 ec80 3400 2760 7150 45f9 eee6  \.)V..4.'qPE...
00000080: fae9 dced 7f2c 8f60 44f8 8636 f3d5 c1ed  ....,`D..6....
00000090: bc9a 5e46 4bf3 cfb7 654c 4a00 1b59 ea81  ..^FK...eLJ..Y..
000000a0: 37f3 7d1f f176 1901 2a54 f9e0 d2e5 955c  7.}...v...*T.....\
000000b0: ed8f 4cfc f8f4 654d 50e9 9253 9764 3eee  ..L...eMP..S.d>..
000000c0: ae82 3a36 4bee 7c26 0022 4e00 2967 e882  ..:6K.|6."N.)g..
000000d0: 28ed 893f e082 4dd8 854a fd84 3400 1a51  (..?..M..J..4...Q
000000e0: 2a3b 5618 304e e9a3 73e8 ae86 ae6c 4100  *;V.ON..s....LA.
000000f0: 266b 0018 4a00 1e41 4036 42ed 7e3b 0017  6k..J..A@6B.~;..
00000100: 5aea a26d dd86 398a 5b41 5f45 42d1 8452  Z..m..9.[A_EB..R
00000110: 282a 41cf 8254 4d3f 4087 6454 061b 336e  (*A..TM?@.dT..3n
00000120: 4f47 ff82 1ba3 7257 b27b 5833 3445 0029  OG.....rW.{X34E.)
00000130: 741d 2541 d67c 39fe 8723 e175 1a7a 5d54  t.%A.|9..#.u.z]T
```

MAP.jpg metadata excerpt (XMP / attribution fields) and AU.png low-level structure (xxd) used during image-side inspection.

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4.2 Fragment Identification and Reconstruction

The four extracted .out files were each identified as JPEG images with identical dimensions, which supports the idea that they were made to act as quarters of a larger map. The provided 2×2 grid layout was used as the reconstruction reference.

Part A / beach.jpg.out

Fragment Name:	Part A (northwest) Auburn map fragment
Source File:	beach.jpg → beach.jpg.out
Extraction Method:	Recovered with stegseek using passphrase ToomersCorner; output verified with file as JPEG image data.
Grid Position:	Part A / Northwest quadrant
Reconstruction Method:	Placed by matching Loachapoka Rd, The Union at Auburn Apartments, Logan Square Apartments, and Martin Luther King Park.

Part B / nyc.jpg.out

Fragment Name:	Part B (northeast) Auburn map fragment
Source File:	nyc.jpg → nyc.jpg.out
Extraction Method:	Recovered with stegseek using passphrase TigerWalk7; output verified with file as JPEG image data.
Grid Position:	Part B / Northeast quadrant
Reconstruction Method:	Placed by matching Bragg Ave, E Glenn Ave, Auburn, Moe's Southwest Grill, and nearby Auburn University references.

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Part C / lena.bmp.out

Fragment Name:	Part C (southwest) Auburn map fragment
Source File:	lena.bmp → lena.bmp.out
Extraction Method:	Recovered with stegseek using passphrase WarEagle2026; output verified with file as JPEG image data.
Grid Position:	Part C / Southwest quadrant
Reconstruction Method:	Placed by matching Auburn University Raptor Center, W Samford Ave, Hemlock Dr, Wire Rd, and AU Campus Recreation and Wellness Center.

Part D / mountain.jpg.out

Fragment Name:	Part D (southeast) Auburn map fragment
Source File:	mountain.jpg → mountain.jpg.out
Extraction Method:	Recovered with stegseek using passphrase Shelby7; output verified with file as JPEG image data.
Grid Position:	Part D / Southeast quadrant
Reconstruction Method:	Placed by matching Jordan-Hare Stadium, Panera Bread, Accolade Auburn, E Samford Ave, and Armstrong St.

Screenshots:

- shows the fully reconstructed Auburn map assembled from the four recovered fragments.
- shows the supplied 2×2 grid used as the reconstruction reference.

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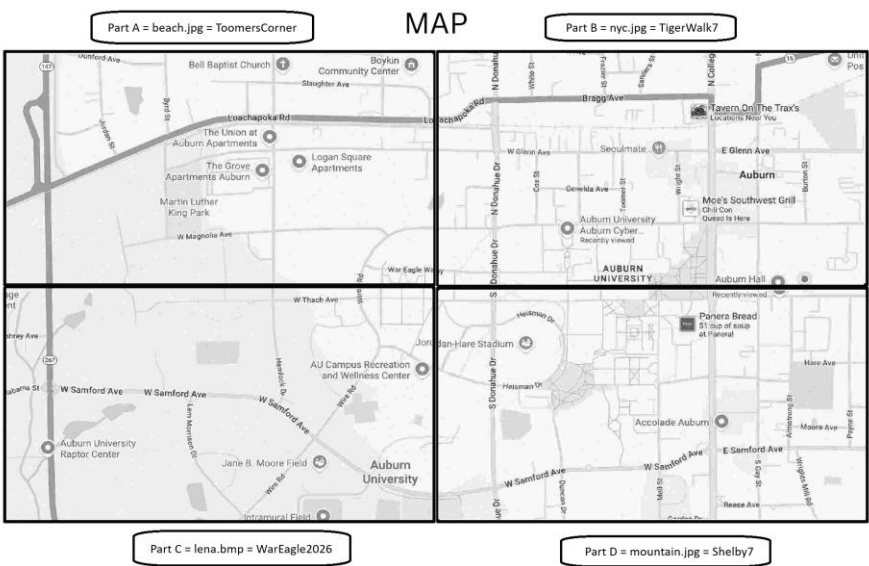


Figure 4.2-1. Reconstructed Auburn map assembled from the four recovered fragment images

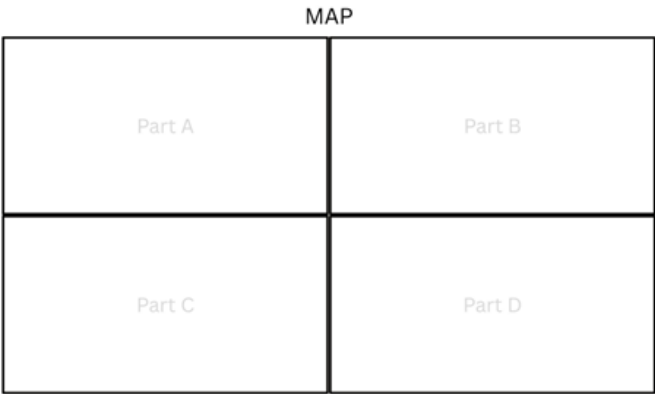


Figure 4.2-2. Supplied 2×2 map layout reference (Parts A–D).

4.3 Decoding of Hidden Messages

Decoded Message Record

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Source (file/fragment):	flag.txt recovered from hidden_archive.zip extracted from SANS_Digital_Forensic.pdf
Encoding Type (binary / octal / hexadecimal):	Hexadecimal text bytes
Tool Used:	cat, xxd -r -p (hex to ASCII conversion)
Decoding Steps:	1) Open flag.txt from the unlocked archive. 2) Observe that the content is stored as hex bytes. 3) Convert the hex bytes to ASCII with xxd -r -p. 4) Interpret the recovered sentence as the meetup instruction.
Decoded Output:	“Lets gather up at Aurelia bank, Auburn at 12 noon. Lets go team!”

After the hidden archive was extracted, flag.txt did not display as plain English at first., it appeared as hexadecimal text bytes. Those bytes were then decoded from hex to ASCII, which had the operational meetup message below.

Screenshots:

Figure 4.3-1 shows flag.txt as hexadecimal text inside the unlocked archive.

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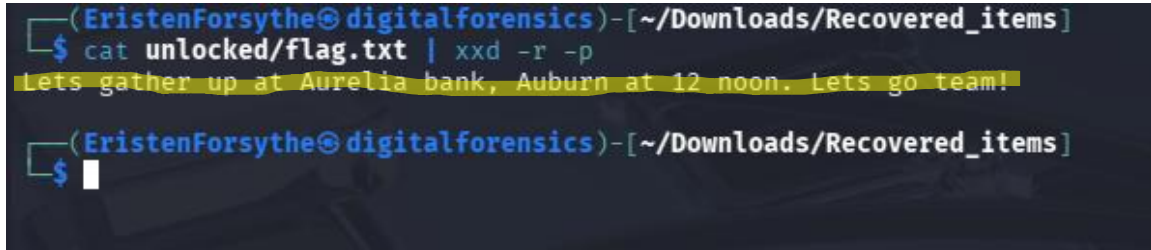
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The shaded command block below records the decode command and the resulting ASCII output.



```
Session Actions Edit View Help
(EristenForsythe@digitalforensics)~[~/Downloads/Recovered_items]
$ cat unlocked/flag.txt
4C 65 74 73 20 67 61 74 68 65 72 20 75 70 20 61 74 20 41 75 72 65 6C 69 61 20 62 61 6E 68 2C 20 41 75 62 75 72 6E 20 61 74
20 31 32 20 6E 6F 6F 6E 2E 20 4C 65 74 73 20 67 6F 20 74 65 61 6D 21
(EristenForsythe@digitalforensics)~[~/Downloads/Recovered_items]
$
```

Figure 4.3-1. flag.txt displayed as hexadecimal text inside the unlocked archive



```
(EristenForsythe@digitalforensics)~[~/Downloads/Recovered_items]
$ cat unlocked/flag.txt | xxd -r -p
Lets gather up at Aurelia bank, Auburn at 12 noon. Lets go team!
(EristenForsythe@digitalforensics)~[~/Downloads/Recovered_items]
$
```

This decoded sentence identifies the target city, the meetup location, and the time of the event. The message is one of the most important findings in the case

4.4 Document and Archive Analysis

SANS_Digital_Forensic.pdf was the most significant document in the evidence set. strings, xxd, and binwalk all pointed to ZIP archive content appended to the PDF. Because the archive was encrypted and automated extraction was incomplete, the archive

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had to be carved manually with dd, verified with file/unzip -l, and then unlocked with the password auburn2026

SANS_Digital_Forensic.pdf

File Name:	SANS_Digital_Forensic.pdf
File Type:	PDF document, version 1.6
Tool Used:	file, strings, xxd, binwalk, dd
Comm and Used:	file /home/EristenForsythe/Downloads/Recovered_items/Recovered_Documents/recovered_documents/SANS_Digital_Forensic.pdf strings /home/EristenForsythe/Downloads/Recovered_items/Recovered_Documents/recovered_documents/SANS_Digital_Forensic.pdf tail -80 xxd /home/EristenForsythe/Downloads/Recovered_items/Recovered_Documents/recovered_documents/SANS_Digital_Forensic.pdf tail -40 binwalk /home/EristenForsythe/Downloads/Recovered_items/Recovered_Documents/recovered_documents/SANS_Digital_Forensic.pdf
Detect ed Hidde n Conte nt:	Appended encrypted ZIP archive containing Daytona_116506_Platinum_Platona_Baguette.jpg, file.pdf, Hope_Diamond.jpg, EP68BLZ.jpg, and flag.txt.

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Extraction Method:	binwalk identified the ZIP entries; dd carved the archive beginning at offset 1486221.
Password Used (if applicable):	The appended archive itself was later unlocked with auburn2026 after carving.
Extracted Files:	hidden_archive.zip (carved archive containing five encrypted members).

hidden_archive.zip

File Name:	hidden_archive.zip
File Type:	ZIP archive data
Tool Used:	dd, file, unzip -l, unzip -P
Command Used:	dd if=/home/EristenForsythe/Downloads/Recovered_items/Recovered_Documents/recovered_documents/SANS_Digital_Forensic.pdf of=hidden_archive.zip bs=1 skip=1486221 file hidden_archive.zip unzip -l hidden_archive.zip unzip -P auburn2026 hidden_archive.zip -d unlocked
Detected Hidden	Five encrypted members were listed inside the ZIP: Daytona image, file.pdf, Hope_Diamond.jpg, EP68BLZ.jpg, and flag.txt.

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Content:	
Extraction Method:	The archive was manually carved from the PDF and then extracted with unzip after password verification.
Password Used (if applicable):	auburn2026
Extracted Files:	Daytona_116506_Platinum_Platona_Baguette.jpg; file.pdf; Hope_Diamond.jpg; EP68BLZ.jpg; flag.txt.

file.pdf

File Name:	file.pdf
File Type:	Composite Document File V2 / OLE-style document (misleading PDF extension)
Tool Used:	file, strings, manual review / LibreOffice
Command Used:	file unlocked/file.pdf strings unlocked/file.pdf head -100
Detected Hidden Content:	Worksheet names and role labels indicating a roster / profit-share record.
Extraction Method:	Recovered from hidden_archive.zip after successful extraction with auburn2026.

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Password Used (if applicable):	auburn2026 (via the archive that contained the file)
Extracted Files:	Not applicable — this file itself contained the roster data.

BAFLE-DCT.pdf

File Name:	BAFLE-DCT.pdf
File Type:	PDF document, version 1.7, 10 pages
Tool Used:	file, directory review
Command Used:	file /home/EristenForsythe/Downloads/Recovered_items/Recovered_Documents/recovered_documents/BAFLE-DCT.pdf
Detected Hidden Content:	No decisive concealed archive or operational artifact was recovered from this file during the documented workflow.
Extraction Method:	The file was inventoried and reviewed as context because its title appeared steganography-related.
Password Used (if applicable):	Not applicable

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Extracted Files:	None recovered during the documented workflow.
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Recovered roster and profit-share evidence from file.pdf:

Name	Role	Profit Share
Cipher	Leader	40%
Phantom	Safe Cracker	25%
Byte	Getaway Driver	15%
Sentinel	Lookout	20%

Screenshots:

- shows manual carving of hidden_archive.zip with dd and verification using file / unzip -l.
- shows binwalk confirming an appended encrypted ZIP archive in SANS_Digital_Forensic.pdf.
- shows extraction of hidden_archive.zip with auburn2026 and the recovered file listing.
- shows file.pdf identified as an OLE-style document with strings exposing the roster labels.
- shows the recovered operational artifacts from the unlocked archive.

```
(EristenForsythe@digitalforensics)~[~/Downloads/Recovered_items]
$ dd if=/home/EristenForsythe/Downloads/Recovered_items/Recovered_Documents/recovered_documents/SANS_Digital_Forensic.pdf of=hidden_archive.zip bs=1 skip=1486221
1846104+0 records in
1846104+0 records out
1846104 bytes (1.8 MB, 1.8 MiB) copied, 4.94587 s, 373 kB/s

(EristenForsythe@digitalforensics)~[~/Downloads/Recovered_items]
$ file hidden_archive.zip
hidden_archive.zip: Zip archive data, made by v3.0 UNIX, extract using at least v2.0, last modified Nov 12 2025 14:15:44,
uncompressed size 326590, method=deflate
Archive: hidden_archive.zip
  Length      Date    Time    Name
  ----      -
  326590    2025-11-12  14:15  Daytona_116586_Platinum_Platona_Baguette.jpg
    6144    2026-03-09  21:28  file.pdf
 1455168    2026-03-08  13:29  Hope_Diamond.jpg
   98312    2025-11-12  14:14  EP68BLZ.jpg
    192     2026-03-09  21:25  flag.txt
 1886406
 5 files
```

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Manual carving of hidden_archive.zip with dd and verification using file and unzip -l.

```
(EristenForsythe@digitalforensics)-[~/Downloads/Recovered_items]
$ binwalk /home/EristenForsythe/Downloads/Recovered_items/Recovered_Documents/recovered_documents/SANS_Digital_Forensic.pdf

DECIMAL      HEXADECIMAL  DESCRIPTION
-----
0            0x0         PDF document, version: "1.6"
330440      0x50AC8     Copyright string: "copyright)/Descent -350/Flags 32/FontBBox[-195 -350 1360 1050]/FontFamily (Fira Sans SemiBold)/FontFile3 133 0 R/FontName/YWKJLF+"
1486221     0x16AD8D    Zip archive data, encrypted at least v2.0 to extract, compressed size: 314292, uncompressed size: 326590, name: Daytona_116506_Platinum_Platona_Baguette.jpg
1800631     0x1B7987    Zip archive data, encrypted at least v2.0 to extract, compressed size: 1401, uncompressed size: 6144, name: file.pdf
1802114     0x1B7F82    Zip archive data, encrypted at least v2.0 to extract, compressed size: 1431291, uncompressed size: 1455168, name: Hope_Diamond.jpg
3233495     0x3156D7    Zip archive data, encrypted at least v2.0 to extract, compressed size: 98098, uncompressed size: 98312, name: EP688LZ.jpg
3331678     0x32D65E    Zip archive data, encrypted at least v2.0 to extract, compressed size: 106, uncompressed size: 192, name: flag.txt
3332303     0x32D8CF    End of Zip archive, footer length: 22

(EristenForsythe@digitalforensics)-[~/Downloads/Recovered_items]
$
```

binwalk output confirming an appended encrypted ZIP archive inside SANS_Digital_Forensic.pdf.

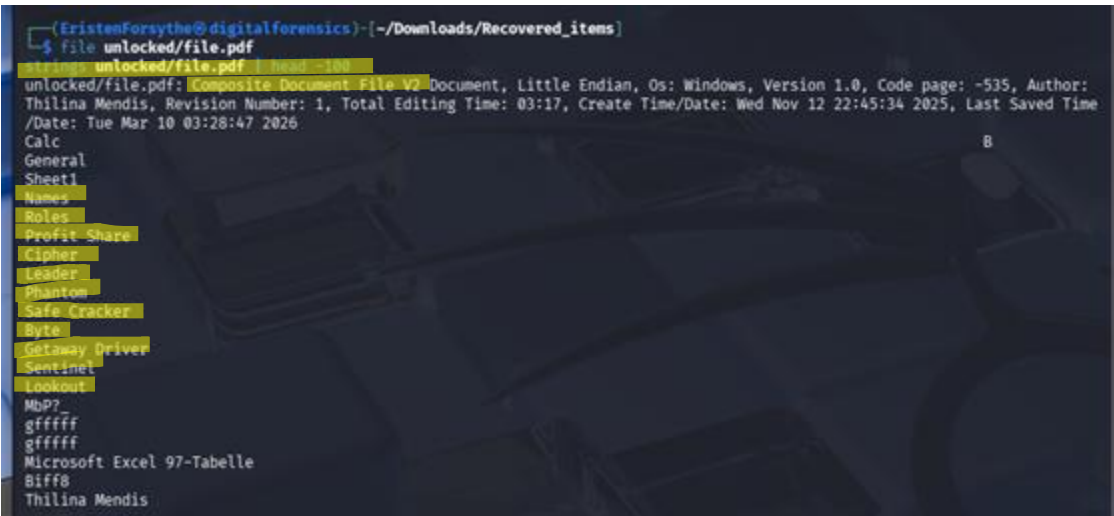
```
(EristenForsythe@digitalforensics)-[~/Downloads/Recovered_items]
$ unzip -p auburn2026 hidden_archive.zip -d unlocked
Archive: hidden_archive.zip
replace unlocked/Daytona_116506_Platinum_Platona_Baguette.jpg? [y]es, [n]o, [A]ll, [N]one, [r]ename: n
replace unlocked/file.pdf? [y]es, [n]o, [A]ll, [N]one, [r]ename: n
replace unlocked/Hope_Diamond.jpg? [y]es, [n]o, [A]ll, [N]one, [r]ename: n
replace unlocked/EP688LZ.jpg? [y]es, [n]o, [A]ll, [N]one, [r]ename: n
replace unlocked/flag.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: n

(EristenForsythe@digitalforensics)-[~/Downloads/Recovered_items]
$ find unlocked -type f
unlocked/Daytona_116506_Platinum_Platona_Baguette.jpg
unlocked/Hope_Diamond.jpg
unlocked/file.pdf
unlocked/EP688LZ.jpg
unlocked/flag.txt
```

Extraction of hidden_archive.zip using auburn2026 and listing of files recovered to the unlocked directory.

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file.pdf identified as an OLE-style document rather than a true PDF, with strings output revealing role labels and worksheet names

Recovered archive file	Recovered with password	Interpretation
flag.txt	auburn2026	Hex-encoded operational meetup message: Aurelia Bank, Auburn, 12 noon.
file.pdf	auburn2026	Misleading extension; identified as an OLE/Excel-style document containing names, roles, and profit shares
EP68BLZ.jpg	auburn2026	Photograph of the green getaway vehicle and its plate EP68 BLZ.
Hope_Diamond.jpg	auburn2026	High-value target-reference image; strongest candidate for the primary target item
Daytona_116506_Platinum Platona_Baguette.jpg	auburn2026	Additional high-value target-reference image of a Rolex Daytona model

The disguised file named file.pdf was particularly important because it contained the team roster, role assignments, and profit split. file recognized it as a composite document/OLE file rather than a normal PDF, and strings exposed worksheet names and role labels.

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Recovered operational artifacts from the unlocked archive: the getaway vehicle, the likely primary target (Hope Diamond), and the additional Rolex target reference

BAFLE-DCT.pdf was also reviewed because it was present in the recovered document directory and appeared steganography related by title. However, the strongest evidence in this case came from the appended archive inside SANS_Digital_Forensic.pdf, not from BAFLE-DCT.pdf.

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5. Conclusion

The evidence shows a clear and well-planned heist. From the images, four passphrases and map fragments were recovered and combined to form a map of Auburn. Further analysis of the documents revealed an encrypted archive hidden inside a PDF, which was unlocked using the password *auburn2026*. This archive contained a decoded message, a hidden roster file, and images related to the plan.

The decoded message revealed that the group planned to meet at Aurelia Bank in Auburn at 12 noon. The roster identified the team members (Cipher, Phantom, Byte, and Sentinel) along with their roles and profit shares. Additional evidence showed a green Ford Mustang (license plate EP68 BLZ) as the getaway vehicle and high-value targets like the Hope Diamond and a Rolex Daytona watch. Overall, the use of steganography, hidden archives, and password protection shows a clear attempt to conceal the operation.

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Extra Proof

The screenshots below preserve additional proof of the investigation workflow. They supplement the main body and show the same workflow in more granular command order

```
(EristenForsythe@digitalforensics)-[~/Downloads/Recovered_items]
$ ls -l
total 148324
-rw-rw-r-- 1 EristenForsythe EristenForsythe 139921593 Mar 29 14:51 passphrases.txt
-rw-rw-r-- 1 EristenForsythe EristenForsythe 3746180 Mar 29 14:51 recovered_documents.zip
-rw-rw-r-- 1 EristenForsythe EristenForsythe 8207836 Mar 29 14:51 recovered_images.zip
(EristenForsythe@digitalforensics)-[~/Downloads/Recovered_items]
$
```

Initial working directory listing showing the staged evidence files.

```
(EristenForsythe@digitalforensics)-[~/Downloads/Recovered_items]
$ unzip recovered_images.zip -d Recovered_Images
unzip recovered_documents.zip -d Recovered_Documents
unzip passphrases.zip -d Passphrases
Archive: recovered_images.zip
  inflating: Recovered_Images/recovered_images/MAP.jpg
  inflating: Recovered_Images/recovered_images/nyc.jpg
  inflating: Recovered_Images/recovered_images/AU.png
  inflating: Recovered_Images/recovered_images/beach.jpg
  inflating: Recovered_Images/recovered_images/mountain.jpg
  inflating: Recovered_Images/recovered_images/Lena.bmp
Archive: recovered_documents.zip
  inflating: Recovered_Documents/recovered_documents/SANS_Digital_Forensic.pdf
  inflating: Recovered_Documents/recovered_documents/BAFLE-DCT.pdf
unzip: cannot find or open passphrases.zip, passphrases.zip.zip or passphrases.zip.ZIP.

(EristenForsythe@digitalforensics)-[~/Downloads/Recovered_items]
$ find /home/EristenForsythe/Downloads/Recovered_items -type f
/home/EristenForsythe/Downloads/Recovered_items/passphrases.txt
/home/EristenForsythe/Downloads/Recovered_items/recovered_images.zip
/home/EristenForsythe/Downloads/Recovered_items/Recovered_Documents/recovered_documents/BAFLE-DCT.pdf
/home/EristenForsythe/Downloads/Recovered_items/Recovered_Documents/recovered_documents/SANS_Digital_Forensic.pdf
/home/EristenForsythe/Downloads/Recovered_items/recovered_documents.zip
/home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/Lena.bmp
/home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/nyc.jpg
/home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/AU.png
/home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/mountain.jpg
/home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/MAP.jpg
/home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/beach.jpg
(EristenForsythe@digitalforensics)-[~/Downloads/Recovered_items]
$
```

Evidence extraction into working folders and file enumeration with find.

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Student Name: Eristen Forsythe

```
(EristenForsythe@digitalforensics): ~/Downloads/Recovered_items
$ find /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images -type f -exec sha256sum {} \;
find /home/EristenForsythe/Downloads/Recovered_items/Recovered_Documents -type f -exec sha256sum {} \;
find /home/EristenForsythe/Downloads/Recovered_items/Passphrases -type f -exec sha256sum {} \;
96bc5194cf541ca8d7537738d11b453fb412678753a40d09d01b021e660a1 /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/lena.bmp
dd318f58c44e4d6b33907e8507ba2225f909b12a9c6b76f3100d1059232db /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/ryce.jpg
6d9438a8c03115fc95eefed959e0b6f7b8905aae7018a38a7376775c83004 /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/AU.png
a70de3b24740dc8dd40b6d1b99f4bf0292db747506088b766c2a927a06aaf01a /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/mountain.jpg
96bc5194cf541ca8d7537738d11b453fb412678753a40d09d01b021e660a1 /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/BAP.jpg
0702b1633a0b0ef4b4f688d4d52852eab0b0e2e942e0eb0b9e4da0d33111d1 /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/beach.jpg
dc8b8c1bba33b1077ce3da05b70a647077ef9cacecf341f462ebf8a08174 /home/EristenForsythe/Downloads/Recovered_items/Recovered_Documents/recovered_documents/recovered_documents/BAPT-DCT.pdf
80711312bc19c9cc0a3683dc99f66d1cee28038b7ae306ffe3782ff407079a3f /home/EristenForsythe/Downloads/Recovered_items/Recovered_Documents/recovered_documents/SANS_Digital_Forensic.pdf
find: '/home/EristenForsythe/Downloads/Recovered_items/Passphrases': No such file or directory
(EristenForsythe@digitalforensics): ~/Downloads/Recovered_items
```

SHA-256 hashes generated for the recovered image, PDF, and passphrases.txt evidence.

```
(EristenForsythe@digitalforensics): ~/Downloads/Recovered_items
$ exiftool /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/*
===== /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/AU.png
ExifTool Version Number      : 13.44
File Name                    : AU.png
Directory                    : /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images
File Size                    : 7.6 kB
File Modification Date/Time  : 2026:03:08 11:09:15-05:00
File Access Date/Time       : 2026:03:29 15:04:20-05:00
File Inode Change Date/Time  : 2026:03:29 15:01:58-05:00
File Permissions             : -rw-rw-r--
File Type                    : PNG
File Type Extension          : png
MIME Type                    : image/png
Image Width                  : 225
Image Height                 : 225
Bit Depth                    : 8
Color Type                   : Palette
Compression                  : Deflate/Inflate
Filter                       : Adaptive
Interlace                   : Noninterlaced
Palette                      : (Binary data 336 bytes, use -b option to extract)
Image Size                   : 225x225
Megapixels                   : 0.051
===== /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/beach.jpg
ExifTool Version Number      : 13.44
File Name                    : beach.jpg
Directory                    : /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images
File Size                    : 483 kB
File Modification Date/Time  : 2026:03:08 12:33:30-05:00
File Access Date/Time       : 2026:03:29 15:04:20-05:00
File Inode Change Date/Time  : 2026:03:29 15:01:58-05:00
File Permissions             : -rw-rw-r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                 : 1.01
Resolution Unit              : inches
X Resolution                  : 72
Y Resolution                  : 72
Image Width                  : 2560
Image Height                 : 1920
Encoding Process             : Baseline DCT, Huffman coding
Bits Per Sample              : 8
Color Components              : 3
Y Cb Cr Sub Sampling         : YCbCr4:2:2 (2 1)
Image Size                   : 2560x1920
Megapixels                   : 4.9
===== /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/lena.bmp
ExifTool Version Number      : 13.44
File Name                    : lena.bmp
Directory                    : /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images
File Size                    : 263 kB
File Modification Date/Time  : 2026:03:08 12:32:05-05:00
File Access Date/Time       : 2026:03:29 15:04:20-05:00
File Inode Change Date/Time  : 2026:03:29 15:01:58-05:00
File Permissions             : -rw-rw-r--
File Type                    : BMP
File Type Extension          : bmp
MIME Type                    : image/bmp
BMP Version                  : Windows V3
Image Width                  : 512
Image Height                 : 512
Planes                       : 1
Bit Depth                    : 8
Compression                  : None
Image Length                 : 262144
Pixels Per Meter X           : 0
Pixels Per Meter Y           : 0
```

exiftool output for recovered image files (page 1 of metadata review).

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```
Num Colors : 256
Num Important Colors : All
Image Size : 512x512
Megapixels : 0.262
===== /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/Map.jpg
ExifTool Version Number : 13.44
File Name : MAP.jpg
Directory : /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images
File Size : 102 kB
File Modification Date/Time : 2025:11:12 11:04:18-06:00
File Access Date/Time : 2026:03:29 15:04:20-05:00
File Inode Change Date/Time : 2026:03:29 15:01:58-05:00
File Permissions : -rw-rw-r--
File Type : JPEG
File Type Extension : jpg
MIME Type : image/jpeg
Exif Byte Order : Little-endian (Intel, II)
Orientation : Horizontal (normal)
X Resolution : 96
Y Resolution : 96
Resolution Unit : inches
Y Cb Cr Positioning : Centered
Exif Version : 0210
Components Configuration : Y, Cb, Cr, -
Flashpix Version : 0100
Color Space : Uncalibrated
Exif Image Width : 2456
Exif Image Height : 1392
Ads Created : 2025-11-12
Ads Ext Id : 910c98c0-48b3-4603-8634-f91dab8d12ba
Ads Fb Id : 525265914179580
Ads Touch Type : 2
Title : Untitled design - 1
Image Width : 2456
Image Height : 1392
Encoding Process : Baseline DCT, Huffman coding
Bits Per Sample : 8
Color Components : 3
Y Cb Cr Sub Sampling : YCbCr4:2:0 (2 2)
Image Size : 2456x1392
Megapixels : 3.4
===== /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/mountain.jpg
ExifTool Version Number : 13.44
File Name : mountain.jpg
Directory : /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images
File Size : 1796 kB
File Modification Date/Time : 2026:03:08 12:34:10-05:00
File Access Date/Time : 2026:03:29 15:04:20-05:00
File Inode Change Date/Time : 2026:03:29 15:01:58-05:00
File Permissions : -rw-rw-r--
File Type : JPEG
File Type Extension : jpg
MIME Type : image/jpeg
JFIF Version : 1.02
Resolution Unit : inches
X Resolution : 72
Y Resolution : 72
Image Width : 4872
Image Height : 3248
Encoding Process : Baseline DCT, Huffman coding
Bits Per Sample : 8
Color Components : 3
Y Cb Cr Sub Sampling : YCbCr4:2:0 (2 2)
Image Size : 4872x3248
Megapixels : 15.8
===== /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images/nyc.jpg
ExifTool Version Number : 13.44
File Name : nyc.jpg
Directory : /home/EristenForsythe/Downloads/Recovered_items/Recovered_Images/recovered_images
File Size : 5.6 MB
```

exiftool output for recovered image files (page 2 of metadata review).

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```
(EristenForsythe@digitalforensics) - [~/Downloads/Recovered_items]
$ xdg-open unlocked/EP68BLZ.jpg
xdg-open unlocked/Daytona_116506_Platinum_Platona_Baguette.jpg
xdg-open unlocked/Hope_Diamond.jpg

(EristenForsythe@digitalforensics) - [~/Downloads/Recovered_items]
$
```

Commands used to open the recovered vehicle and target reference images after archive extraction.

images

```
(EristenForsythe@digitalforensics) - [~/Downloads/Recovered_items]
$ file beach.jpg.out lena.bmp.out mountain.jpg.out nyc.jpg.out
beach.jpg.out: JPEG image data, JFIF standard 1.01, resolution (DPI), density 144x144, segment length 16, baseline, precision 8, 1228x696, components 1
lena.bmp.out: JPEG image data, JFIF standard 1.01, resolution (DPI), density 144x144, segment length 16, baseline, precision 8, 1228x696, components 1
mountain.jpg.out: JPEG image data, JFIF standard 1.01, resolution (DPI), density 144x144, segment length 16, baseline, precision 8, 1228x696, components 1
nyc.jpg.out: JPEG image data, JFIF standard 1.01, resolution (DPI), density 144x144, segment length 16, baseline, precision 8, 1228x696, components 1

(EristenForsythe@digitalforensics) - [~/Downloads/Recovered_items]
$
```

File discovery showing the carved archive and extracted evidence artifacts.

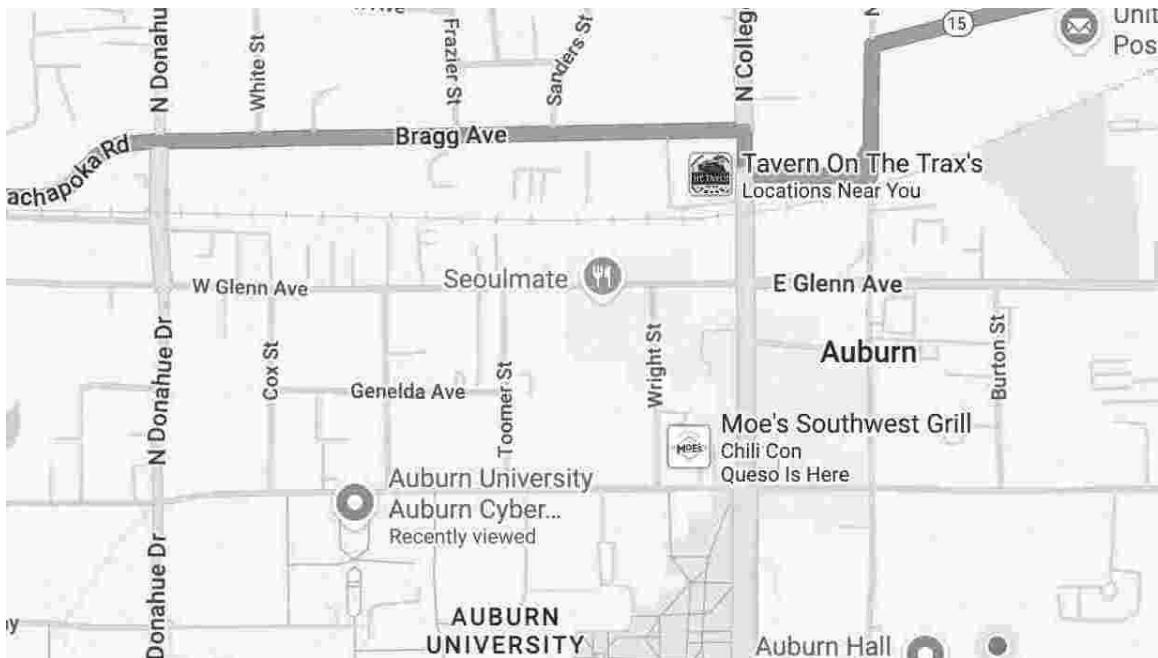
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Recovered image artifacts shown individually for quick visual reference



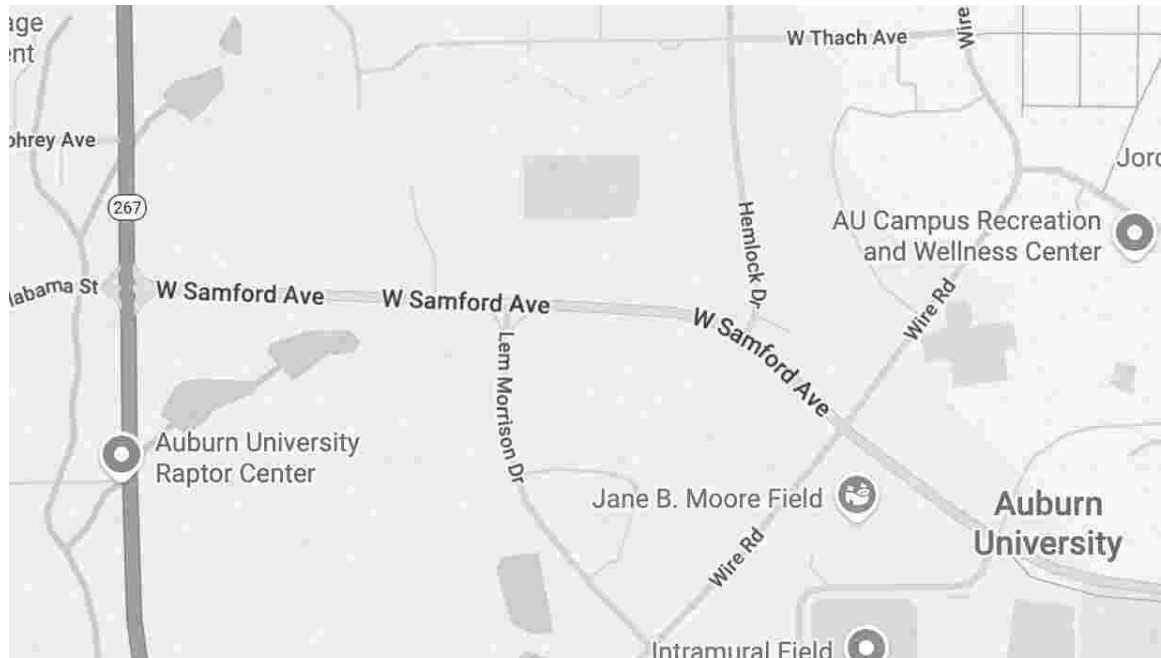
beach.jpg.out fragment (Part A / northwest).



nyc.jpg.out fragment (Part B / northeast).

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lena.bmp.out fragment (Part C / southwest).



mountain.jpg.out fragment (Part D / southeast).

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EP68BLZ.jpg : green Ford Mustang getaway vehicle.



Hope_Diamond.jpg : likely primary target reference.

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Daytona_116506_Platinum_Platona_Baguette.jpg : additional target reference.